



MBARARA UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATICS

BACHELOR OF SCIENCE COMPUTER SCIENCE

CSC1204 DATABASE SYSTEMS

LECTURER: MR. KENNETH BAGUMA

NAME: ASIIMWE GILBERT

REG NUMBER: 2025/BCS/050/PS

PROJECT DOMAIN: OIL AND GAS SECTOR

PROJECT CHOICE: HEALTH, SAFETY AND ENVIRONMENT DBMS

GROUP: GROUP 5

1. Executive Summary

This report details the design, development, and implementation of a Health, Safety, and Environment (HSE) Database Management System tailored for the emerging Oil and Gas sector in Uganda (specifically focusing on operations in the Albertine Graben, such as the Tilenga and Kingfisher projects). As oil exploration and production intensify, managing operational risks, ensuring regulatory compliance with the National Environment Management Authority (NEMA) and the Petroleum Authority of Uganda (PAU), and safeguarding human lives are paramount. The proposed MySQL-based system transitions HSE data management from fragmented, paper-based or spreadsheet-heavy processes into a centralized, relational architecture. This system streamlines incident reporting, tracks employee safety training, and manages environmental audits, ultimately fostering a proactive safety culture, and mitigating regulatory risks.

2. Business Case

a. Application Domain:

The HSE Database System is designed to support core safety and environmental operations within an oil exploration and production company. Key processes supported include:

- Incident Management: Logging workplace accidents, near-misses, and hazardous observations.
- Audit Tracking: Scheduling and recording environmental and safety inspections at various rig and camp sites.
- Compliance and Training: Tracking employee safety certifications (e.g., BOSIET, First Aid) to ensure field readiness.
- User Groups Served: HSE Managers (strategic oversight, reporting), Field Safety Officers (data entry, incident logging), HR Coordinators (training updates), and external compliance auditors (read-only access).

b. Analysis of Benefits

- **Tangible Benefits:** Reduction in regulatory fines from PAU/NEMA through automated compliance tracking; lower insurance premiums due to demonstrable risk management; decreased administrative hours spent compiling monthly safety reports.
- **Intangible Benefits:** Enhanced organizational reputation; improved employee morale and trust due to a visible commitment to safety; facilitation of a proactive rather than reactive safety culture through data-driven insights.

c. Analysis of Costs

The system is economically feasible, with long-term benefits far outweighing initial expenditures.

- **Tangible Costs:** Server hosting/cloud infrastructure for MySQL deployment, development and testing hours, procurement of mobile devices for field data entry, and database maintenance/backup storage.
- **Intangible Costs:** Temporary reduction in operational efficiency during the system adoption and user onboarding phases.

d. User Issues

A primary challenge is the varying levels of digital literacy among field workers and site contractors in remote Ugandan oil fields. Adoption effort will require intuitive, localized user interfaces (potentially supporting Swahili or Luganda tooltips) and comprehensive, hands-on training sessions to ensure accurate data entry at the source.

e. Risks Assessment

- **Technical Risks:** Intermittent internet connectivity at remote drilling sites (e.g., Buliisa, Hoima) causing data synchronization delays.
- **Organizational Risks:** Resistance to change personnel accustomed to legacy reporting methods; under-reporting of incidents due to fear of punitive action (requires policy-level mitigation).
- **Security Risks:** Unauthorized access to sensitive medical or incident data, necessitating strict Role-Based Access Control (RBAC).

f. Limitations of the System

The system is fundamentally an information repository and analytical tool; it cannot physically prevent accidents. Its efficacy is entirely dependent on the integrity and timeliness of human data entry ("garbage in, garbage out"). Furthermore, the current iteration does not include offline-first mobile capabilities, requiring an active connection to commit records to the central database.

3. Design: Logical and Physical Design

Logical Design

a. List of Entities

- Site (Operational locations like rigs or camps)
- Employee (Personnel working at the sites)
- Incident (Safety or environmental events)
- Audit (Scheduled site inspections)
- Training (Safety courses)

b. Entities and Attributes

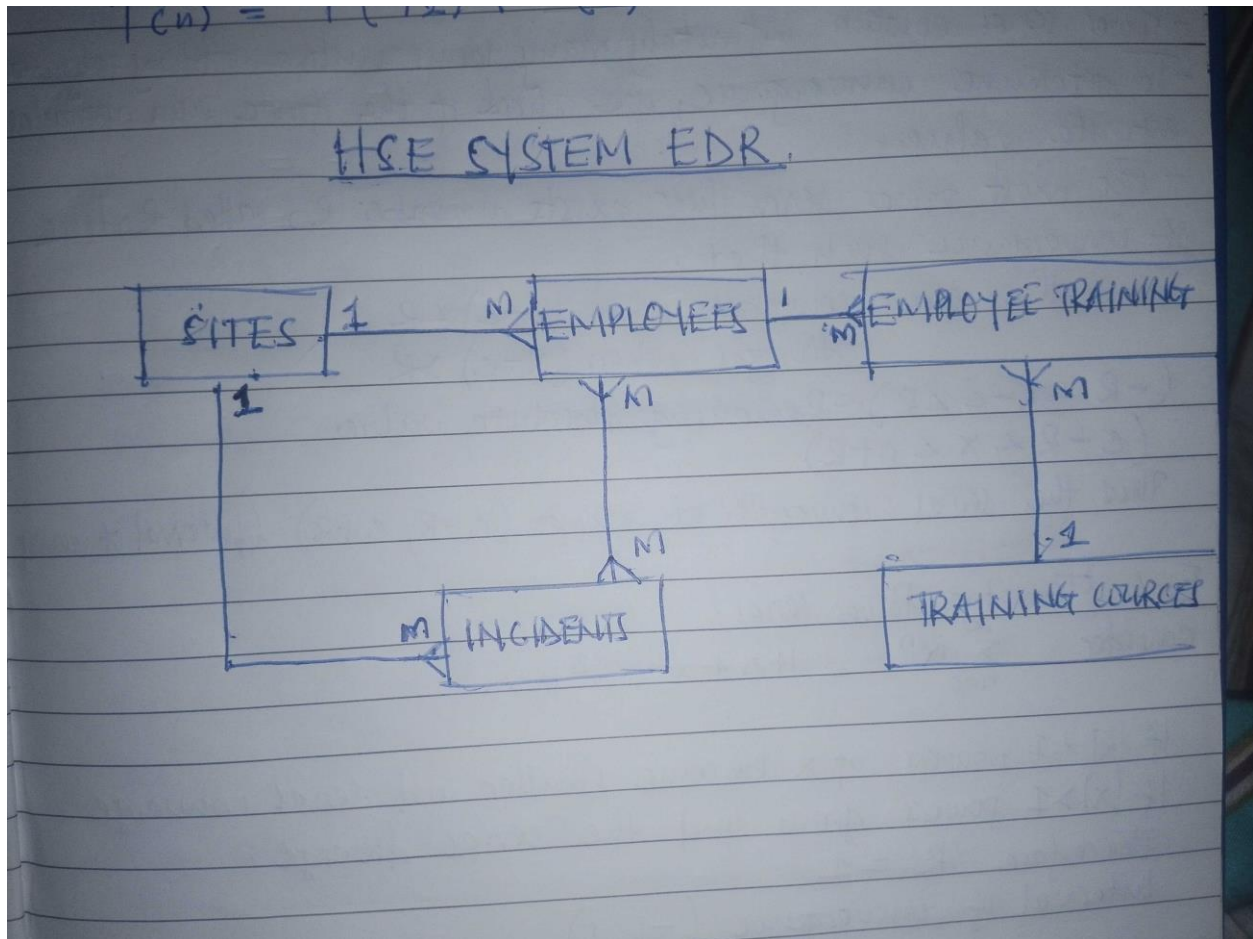
- Site: Site_ID (PK), Site_Name, Location, Region, Risk_Level.
- Employee: Emp_ID (PK), First_Name, Last_Name, Role, Hire_Date, Site_ID (FK).
- Incident: Incident_ID (PK), Date_Time, Category, Description, Severity, Emp_ID (FK), Site_ID (FK).
- Audit: Audit_ID (PK), Audit_Date, Auditor_Name, Score, Remarks, Site_ID (FK).
- Training: Course_ID (PK), Course_Name, Validity_Months.
- Emp_Training (Junction): Emp_ID (PK/FK), Course_ID (PK/FK), Completion_Date, Expiry_Date.

c. Entity Relationships & d. Multiplicities

- A Site hosts many Employees (1:M / One-to-Many).
- An Employee can report/be involved in many Incidents (1:M).
- A Site can have many Incidents occurring at it (1:M).
- A Site undergoes many Audits (1:M).

- An Employee takes many Trainings, and a Training is taken by many Employees (M:N / Many-to-Many), resolved by the Emp_Training junction table.

e. Comprehensive ERD (Conceptual Description)



f. User Interface (Dialogue Structure)

1. Login Screen: Prompts for Username and Password.



HEALTH, SAFETY & ENVIRONMENT SYSTEM

UGANDA OIL & GAS SECTOR · SECURE ACCESS

USERNAME

PASSWORD



Sign In to HSE System

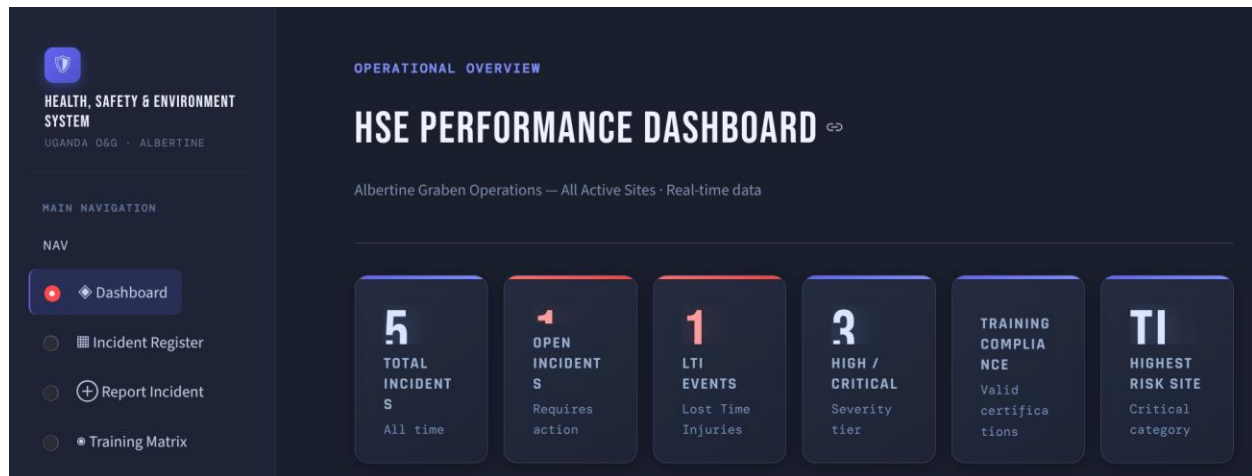
HSE DATABASE MANAGEMENT SYSTEM

Tilenga · Kingfisher · EACOP Operations

Albertine Graben · Uganda

Authorised personnel only · v2.0

2. Main Dashboard: Displays KPI widgets.



3. Report Incident Flow:

- Select Site (Dropdown) -> Select Category (LTI, Near Miss, Spill) -> Enter Description and Severity.
- Submit -> System validates data -> Displays "Incident Successfully Logged" -> Returns to Dashboard.

The screenshot shows the 'REPORT NEW INCIDENT' form. It includes a sidebar with the same navigation as the dashboard, where 'Report Incident' is selected. The main form area has a title 'REPORT NEW INCIDENT' and a note: 'All fields marked * are mandatory. Records are committed to the HSE database immediately.' The form is divided into two sections: 'INCIDENT DETAILS' and 'CLASSIFICATION'. The 'INCIDENT DETAILS' section contains the following fields:

- INCIDENT DATE ***: 2026/05/06
- INCIDENT TIME ***: 08:00
- SITE ***: Buhimba Logistics Base (dropdown)
- FIELD SAFETY OFFICER (REPORTER) ***: Harriet Akello — Driller (dropdown)

The 'CLASSIFICATION' section is partially visible at the bottom.

Physical Design

g. Database Schema (MySQL Layout)

- **sites** (site_id INT AUTO_INCREMENT PK, site_name VARCHAR(50), location VARCHAR(50), risk_level ENUM('Low', 'Medium', 'High'))

- **employees** (emp_id INT AUTO_INCREMENT PK, f_name VARCHAR(50), l_name VARCHAR(50), role VARCHAR(50), site_id INT FK)
- **incidents** (incident_id INT AUTO_INCREMENT PK, incident_date DATE, category VARCHAR(50), severity ENUM('Minor', 'Major', 'Critical'), emp_id INT FK, site_id INT FK)
- **trainings** (course_id INT AUTO_INCREMENT PK, course_name VARCHAR(100), validity_months INT)
- **emp_training** (emp_id INT FK, course_id INT FK, completion_date DATE, expiry_date DATE, PRIMARY KEY(emp_id, course_id))

h. Description of Queries and Reports

- **Incident Frequency Report:** Aggregates incidents by site and severity. Business Need: Allows HSE managers to identify high-risk locations (e.g., Kingfisher Rig 2) needing immediate intervention.
- **Training Compliance Report:** Identifies employees whose critical safety certifications expire within 30 days. Business Need: Ensures regulatory compliance and prevents unauthorized personnel from entering high-risk zones.

4. Implementation

Note: The following represents the actual MySQL code and simulated output (screen representations) for the system.

Table Creation (DDL)

```
DROP DATABASE IF EXISTS hse_db;
CREATE DATABASE hse_db
  CHARACTER SET utf8mb4
  COLLATE      utf8mb4_unicode_ci;

USE hse_db;
```

```
-- TABLE: sites
-----
CREATE TABLE sites (
  site_id          INT          NOT NULL AUTO_INCREMENT,
  site_name        VARCHAR(100) NOT NULL,
  site_type        ENUM(
    'Drilling Pad',
    'Flow Station',
    'Pipeline Corridor',
    'Logistics Base',
    'Camp'
  ) NOT NULL,
  district         VARCHAR(50)  NOT NULL,
  region           VARCHAR(50)  NOT NULL DEFAULT 'Albertine Graben',
  gps_lat          DECIMAL(9,6) NULL,
  gps_lng          DECIMAL(9,6) NULL,
  hse_risk_category ENUM('Low', 'Medium', 'High', 'Critical')
    NOT NULL DEFAULT 'Medium',
  commissioned_date DATE        NULL,
  site_status      ENUM('Operational', 'Standby', 'Decommissioned')
    NOT NULL DEFAULT 'Operational',
  PRIMARY KEY (site_id),
  INDEX idx_district (district),
  INDEX idx_type     (site_type)
) ENGINE=InnoDB;
```

```
-- TABLE: incidents
-----
CREATE TABLE incidents (
  incident_id      INT          NOT NULL AUTO_INCREMENT,
  incident_date    DATETIME    NOT NULL,
  site_id          INT          NOT NULL,
  reported_by      INT          NOT NULL,
  involved_employee_id INT      NULL,
  incident_type    ENUM(
    'Near-Miss',
    'First Aid',
    'MTC',
    'LTI',
    'Fatality',
    'Environmental'
  ) NOT NULL,
  severity         ENUM('Low', 'Medium', 'High', 'Critical')
    NOT NULL DEFAULT 'Medium',
  description      TEXT        NOT NULL,
  root_cause       ENUM(
    'Procedural Violation',
    'Equipment Failure',
    'Human Error',
    'Environmental Condition',
    'Management System Gap'
  ) NOT NULL
);
```

```
-- TABLE: employees
-- -----
CREATE TABLE employees (
  employee_id      INT          NOT NULL AUTO_INCREMENT,
  first_name       VARCHAR(50)  NOT NULL,
  last_name        VARCHAR(50)  NOT NULL,
  national_id      VARCHAR(20)  NOT NULL,
  job_title        VARCHAR(100) NOT NULL,
  department       VARCHAR(50)  NOT NULL,
  site_id         INT          NOT NULL,
  employment_type  ENUM('Direct','Contractor','Sub-Contractor')
                  NOT NULL DEFAULT 'Direct',
  date_of_hire     DATE         NOT NULL,
  contact_phone    VARCHAR(20)  NULL,
  emp_status       ENUM('Active','On Leave','Terminated')
                  NOT NULL DEFAULT 'Active',
  PRIMARY KEY (employee_id),
  UNIQUE KEY uq_national_id (national_id),
  FOREIGN KEY fk_emp_site (site_id)
    REFERENCES sites(site_id)
    ON UPDATE CASCADE
    ON DELETE RESTRICT,
  INDEX idx_emp_status (emp_status),
  INDEX idx_emp_dept  (department)
) ENGINE=InnoDB;
```

Populating the Database (DML)

```
-- =====
-- DML: SAMPLE DATA
-- =====

INSERT INTO sites (site_name, site_type, district, region, gps_lat, gps_lng, hse_risk)
('Tilenga Pad TP-09',      'Drilling Pad',      'Buliisa',  'Albertine Graben',  1.
('Tilenga Pad TP-14',      'Drilling Pad',      'Buliisa',  'Albertine Graben',  1.
('Kingfisher Flow Station', 'Flow Station',      'Kikuube',  'Albertine Graben',  1.
('Buhimba Logistics Base', 'Logistics Base',    'Hoima',    'Albertine Graben',  1.
('EACOP KP-045 Corridor',  'Pipeline Corridor', 'Kikuube',  'Albertine Graben',  1.

INSERT INTO employees (first_name, last_name, national_id, job_title, department, site
('Amos',      'Kaboyo',      'UG-CM-2019-0012', 'HSE Supervisor',      'HSE',
('Grace',     'Namutebi',     'UG-KLA-2020-0445', 'Field Safety Officer', 'HSE',
('Patrick',   'Okello',      'UG-GL-2018-0338', 'Rig Manager',          'Drilling',
('Harriet',   'Akello',      'UG-BU-2021-0097', 'Driller',               'Drilling',
('Emmanuel',  'Byaruhanga',  'UG-KIK-2020-0221', 'Field Safety Officer', 'HSE',
('Judith',    'Nassimbwa',   'UG-HOI-2022-0088', 'Logistics Coordinator', 'Logistics',
('Robert',    'Tugume',      'UG-BU-2019-0501', 'Pipeline Inspector',   'Engineering',
('Christine', 'Atim',        'UG-GL-2023-0003', 'Rig Hand',              'Drilling',
('Moses',     'Waiswa',      'UG-KLA-2017-0200', 'Senior HSE Auditor',   'HSE',
('Florence',  'Amoding',     'UG-KIK-2021-0155', 'Environmental Officer', 'HSE',
```

```
INSERT INTO training_courses (course_name, course_code, category, validity_months, is_
('BOSIET – Basic Offshore Safety Induction & Emergency Training', 'BOSIET', 'Safety
('H2S Awareness and Breathing Apparatus', 'H2S-BA', 'Hazard
('IWCF Well Control – Level 3 (Drilling)', 'IWCF-L3', 'Well C
('First Aid Level 3 – Occupational', 'FA-L3', 'First
('NEBOSH International General Certificate', 'NEBOSH-IGC', 'Leader
('Environmental Impact Assessment Practitioner', 'EIA-PRAC', 'Enviro
('Fire Warden & Emergency Response', 'FWER', 'Emerge
('Pipeline Integrity Management (ASME B31.8)', 'PIM-B318', 'Safety

INSERT INTO incidents (incident_date, site_id, reported_by, involved_employee_id, inci
('2024-02-10 08:35:00', 1, 2, 3, 'Near-Miss', 'Medium',
'Drill pipe stand improperly racked, nearly fell onto rig floor during pipe-tripping
'Procedural Violation',
'Operations halted. Area cordoned and pipe stand re-racked by competent rigger.',
'Refreshers toolbox talk on pipe-racking procedures. Update SOP DRL-007.',
'Closed', '2024-02-15'),
('2024-03-22 14:10:00', 1, 2, 4, 'First Aid', 'Low',
'Rig hand sustained laceration to left forearm while handling valve manifold. Require
'Human Error',
'First aid administered on-site. Incident communicated to rig manager.',
'Mandatory glove inspection before manifold handling. Update PPE checklist.',
'Closed', '2024-03-28'),
```

Sample Queries and Results (Simulated Screenshots)

Query 1: Generate an Incident Report with Location and Reporting Employee

```
> mysql -u root -p hse_db -e "
SELECT
  i.incident_id AS 'ID',
  i.incident_date AS 'Timestamp',
  s.site_name AS 'Location',
  s.district AS 'District',
  CONCAT(e.first_name, ' ', e.last_name) AS 'Reporting Employee',
  i.incident_type AS 'Type',
  i.severity AS 'Severity'
FROM incidents i
JOIN sites s ON i.site_id = s.site_id
JOIN employees e ON i.reported_by = e.employee_id
ORDER BY i.incident_date DESC;"
Enter password:
```

ID	Timestamp	Location	District	Reporting Employee	Type	Severity
5	2024-09-03 16:45:00	EACOP KP-045 Corridor	Kikuube	Robert Tugume	Near-Miss	Critical
4	2024-07-19 11:22:00	Kingfisher Flow Station	Kikuube	Emmanuel Byaruhanga	Environmental	High
3	2024-05-07 06:50:00	Tilenga Pad TP-14	Buliisa	Grace Namutebi	LTI	High
2	2024-03-22 14:10:00	Tilenga Pad TP-09	Buliisa	Grace Namutebi	First Aid	Low
1	2024-02-10 08:35:00	Tilenga Pad TP-09	Buliisa	Grace Namutebi	Near-Miss	Medium

Query 2: Identify Employees with Expired Training Courses

```

> mysql -u root -p hse_db -e "
SELECT
    CONCAT(e.first_name, ' ', e.last_name) AS 'Employee',
    e.job_title AS 'Role',
    tc.course_name AS 'Training Course',
    et.expiry_date AS 'Expiration Date'
FROM employee_training et
JOIN employees e ON et.employee_id = e.employee_id
JOIN training_courses tc ON et.course_id = tc.course_id
WHERE et.expiry_date < CURDATE()
ORDER BY et.expiry_date ASC;"
Enter password:

```

Employee	Role	Training Course	Expiration Date
Christine Atim	Rig Hand	Fire Warden & Emergency Response	2024-05-03
Grace Namutebi	Field Safety Officer	First Aid Level 3 – Occupational	2024-06-01
Harriet Akello	Driller	H2S Awareness and Breathing Apparatus	2024-07-14
Robert Tugume	Pipeline Inspector	H2S Awareness and Breathing Apparatus	2024-09-10
Grace Namutebi	Field Safety Officer	BOSIET – Basic Offshore Safety Induction & Emergency Training	2024-09-20
Patrick Okello	Rig Manager	IWCF Well Control – Level 3 (Drilling)	2024-11-01
Grace Namutebi	Field Safety Officer	Fire Warden & Emergency Response	2024-11-10
Amos Kaboyo	HSE Supervisor	Fire Warden & Emergency Response	2024-11-10
Judith Nassimbwa	Logistics Coordinator	Fire Warden & Emergency Response	2024-11-20
Amos Kaboyo	HSE Supervisor	BOSIET – Basic Offshore Safety Induction & Emergency Training	2025-01-15
Emmanuel Byaruhanga	Field Safety Officer	Fire Warden & Emergency Response	2025-01-15

5. Bibliography

1. Connolly, T. M., & Begg, C. E. (2014). Database Systems: A Practical Approach to Design, Implementation, and Management (6th ed.). Pearson.
2. Elmasri, R., & Navathe, S. B. (2015). Fundamentals of Database Systems (7th ed.). Pearson.
3. Petroleum Authority of Uganda (PAU). (2023). Health, Safety and Environment (HSE) Regulations and Standards in the Albertine Graben. Government of Uganda Publications.
4. Speegle, G. D. (2001). Developing Client/Server Applications with Oracle Developer/2000. (Adapted principles for relational DB integration in industrial settings).
5. Oracle Corporation. (2023). MySQL 8.0 Reference Manual. Retrieved from dev.mysql.com/doc/refman/8.0/en/